

Name: Tanish

Class: LAB-2VC

Topic: ELEVEN YEARS. Acceleration at its finest

Part		Pages	Sections		History covered	
I. The Late Start		1 to 19 (19)	10		4 Oct 1957 to 5 May 1961 (3 yr 7 mo)	
II. The Deadline		20 to 34 (15)	10		25 May 1961 to 31 Jul 1964 (3 yr 2 mo)	
III. The Bridge		35 to 46 (12)	8		23 Mar 1965 to 15 Nov 1966 (1 yr 8 mo)	
IV. Fire and Reconnaissance		47 to 61 (15)	12		27 Jan 1967 to 30 Jun 1969 (2 yr 5 mo)	
V. Twelve and a Half Minutes		62 to 72 (11)	9		16 to 24 Jul 1969 (8 days) + coda	

Chapter name	Pages	Written Content	In
PART I. THE LATE START. Half-title. The First Question	1	No body text. One question alone on an almost empty page, set at 48/52 in the display face: How long does the trip take, from nowhere to the Moon? Beneath the question, in monospace at 8 pt, the interval the book covers: 1 Oct 1958 to 24 Jul 1969. 10 years, 9 months, 23 days.	
The Whole Eleven Years. Contents and timeline gatefold	2-3	Contents list on p3, set as five parts with date spans and page ranges. The gatefold carries no prose: only event ticks, each with a date in monospace and a four-word caption. Two parallel rows of ticks. The firsts that had already happened, and the answers that came after, sharing one time axis.	
Forty-Three Years Before	4-5	The agency NASA absorbed. NACA was founded in 1915 as a research committee for aeronautics, not a space agency. When NASA opened, NASA took NACA intact: 8,000 employees, an annual budget of US\$100 million, three major laboratories (Langley, Ames, Lewis), and two small test facilities. What NASA did not take: a rocket, a spacecraft, a crew, or a deadline.	
4 October 1957. The Beep Heard First	6-7	The first artificial satellite entered orbit just under twelve months before NASA existed. The transmitter ran for twenty-one days. America was already behind before America had an agency to be behind with.	
1 October 1958. T minus 0	8-9	NASA opens for business. What the new agency had on day one, and what the new agency did not. The eleven years start here. 1 Oct 1958 to 24 Jul 1969 is ten years and nine months. The title rounds up to eleven because the deadline was the decade, and the decade had five months left when the decade was met.	

The Ledger of Being Second	10-11	Every Soviet first from 1957 to 1966 with a date, and the American answer with a date wherever there was one: first satellite, Oct 1957. First living creature in orbit, Nov 1957. First lunar impact, Sep 1959. First images of the far side, Oct 1959. First human in orbit, 12 Apr 1961. First woman in orbit, Jun 1963. First soft lunar landing, Feb 1966. First spacewalk, 18 Mar 1965. The column of American answers is visibly shorter and visibly later.	
9 April 1959. Seven Men, No Rocket	12-13	The first American astronauts were announced in April 1959, chosen from military test pilots. The vehicle to carry them did not exist yet. The flight they were chosen for would last fifteen minutes.	
12 April 1961. One Hundred and Eight Minutes	14-15	The first human orbit of the Earth: one orbit, 108 minutes, and a safe return. Twenty-three days later the United States put a man in space for the first time, and the man did not reach orbit.	
5 May 1961. Fifteen Minutes	16-17	The first American in space flew a suborbital arc of 15 minutes 22 seconds. The flight was real and the flight was not in orbit. The gap between the two is the gap this book is about, and the gap was still open when the deadline arrived twenty days later.	
The Ladder	18-19	Every launch vehicle of the eleven years, described once each in three lines: what the vehicle was derived from, what the vehicle lifted, and what the vehicle had to prove. Redstone, Jupiter-C, Atlas, Titan II, Saturn I, Saturn IB, Saturn V. The last of these had no predecessor at the scale and had to be invented from nothing while the others were still flying.	
PART II. THE DEADLINE. The Question the Ladder Cannot Answer	20	One page, one question, no answer yet: How many of these had to fly before anyone put a person on the next one? The conventional answer in 1961 was every one of them, in order, which is a schedule ending after 1970.	

25 May 1961. Eight Years, Seven Months	21-22	The President asked Congress for a national commitment to landing a man on the Moon and returning the man safely to the Earth before the decade was out. The commitment was announced twenty days after the first American suborbital flight and forty-three days after the first human orbit, at a point when the entire United States record in crewed spaceflight stood at fifteen minutes and twenty-two seconds. The deadline was not set from a position of strength. The deadline was set to create one.
12 September 1962. Rice	23-24	The public restatement, delivered at Rice Stadium in Houston to a crowd commonly cited at around 35,000 on a hot afternoon: We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard. The sentence is quoted in full, with the surrounding paragraph, because the book needs the reader to hear the difficulty stated as the reason and not as an obstacle.
Three Routes, One That Fits	25	How to get there was not settled. Direct ascent, one enormous vehicle straight down to the surface, was the assumption. Earth-orbit rendezvous, built in pieces and assembled in low orbit, was the engineer's preference. Lunar-orbit rendezvous, send a small lander down from lunar orbit and bring only the lander back, was the unfashionable answer, argued from Langley through 1961 and 1962, and resisted because the method meant docking two spacecraft at 240,000 miles from home. Lunar-orbit rendezvous was chosen because the method halved the mass to be landed, and the method was the only one of the three that fit inside eight years.

Three Centres, One Machine	26-27	The work was split so the work happened at once rather than in sequence. Marshall in Huntsville took the launch vehicle. The new Manned Spacecraft Center in Houston took the spacecraft and the crews. The Launch Operations Center in Florida, renamed for Kennedy in 1963, took the pad. Langley, Ames, Lewis, and the Jet Propulsion Laboratory kept the research running underneath. At peak the effort employed about 400,000 people and required the support of more than 20,000 industrial firms and universities. In the mid-1960s the agency itself held roughly 33,200 federal employees alongside some 377,000 contractor staff. Parallelism was not an organisational preference. Parallelism was the speed technique.
September 1963. The Man Who Changed the Plan	28	George Mueller took charge of manned space flight in September 1963 and inherited a schedule unable to reach the Moon by 1970. Mueller rebuilt the management structure so systems were developed concurrently instead of one after another, and Mueller cut the number of test flights, which meant changing what a test flight was for.
The Argument	29-30	The single decision that bought the years. The established method was to fly a new rocket stage by stage: ballast the upper stages, prove the first, then add a live second stage on the next flight. Mueller proposed the very first flight of the largest rocket ever built carrying all three live stages and a live spacecraft. The director of Marshall argued if the flight failed, nobody would say why. Mueller outranked the director, and the plan was adopted. Von Braun's own later account: the plan sounded reckless, but George Mueller's reasoning was impeccable, and without all-up testing the first manned lunar landing would not have taken place as early as 1969. The third Saturn V ever launched carried people to the Moon. The sixth carried the first landing.

The Money Curve	31	Acceleration has a price and the price was paid up front. NASA's budget was US\$100million in 1958 and reached US\$5.933 billion in fiscal year 1966, 4.41% of the entire United States federal budget, the highest share the agency has ever held. Apollo cost US\$25.4billion in the dollars of the day across fiscal years 1960 to 1969, or \$257 billion in 2020 dollars, and accounted for more than half of all NASA spending through the decade.
31 July 1964. Falling at the Moon	32-33	Before anyone was sent, the ground had to be seen. Ranger 7 launched on 28 July 1964 and transmitted 4,308 photographs in the seventeen minutes before the probe destroyed itself on the surface. Images a thousand times clearer than anything any Earth-bound telescope had produced. Ranger 8 followed in February 1965 with more than 7,000 images and crashed into the Sea of Tranquility, four and a half years before Apollo 11 landed there.
The Board at the End of 1964	34	A full-page status board at the three-year mark of the countdown: what existed, what was on the pad, what was on the drawing board, and what did not exist at all. No rocket large enough had flown. No spacecraft had carried three people. Nobody had walked in space. Nobody had rendezvoused with anything. Two years and seven months remained.
PART III. THE BRIDGE. 23 March 1965. Two Seats	35-36	Gemini was not built to reach the Moon. Gemini was built to make reaching the Moon possible: to prove rendezvous, docking, spacewalking, duration, and precision re-entry, in that order, on a converted ballistic missile. Gemini 3 flew 4 hours 52 minutes and made the first manual orbital manoeuvre. Ten crewed flights followed in twenty months, carrying twenty astronauts through 1,104 orbits. Fifteen of the twenty-four people who later flew to the Moon had flown Gemini first.

3 June 1965. Twenty-Three Minutes	37	The first American spacewalk: twenty-three minutes outside the spacecraft, tethered, with no way back in if the hatch jammed. The astronaut did not want to come in and had to be ordered to.
August 1965. Eight Days, Then Fourteen	38-39	Duration was unknown, not a specification. Nobody knew whether a human body worked for eight days in weightlessness, and the Moon trip needed eight. Gemini 5 flew 7 days 22 hours on the first use of fuel cells. Gemini 7 then held 13 days 18 hours in a cabin the size of a car's front seat, and Gemini 7 was used as the target for the first rendezvous.
15 December 1965. One Foot	40	The hardest item on the list, and the one that makes the lunar plan possible: two crewed spacecraft launched a day apart, closing on each other in orbit and holding station about a foot apart for hours. If this was not done 100 miles up, the task was not done 240,000 miles out with a lander and a way home depending on the task.
16 March 1966. The Spin	41-42	The first docking in space, and the first in-flight emergency of the American programme. Minutes after docking with an unmanned target, the combined vehicles began to roll and would not stop. The cause was a stuck thruster on the crewed spacecraft. The crew undocked, fought a violent tumble, and aborted the mission after 10 hours 41 minutes. The pilot who flew the vehicle out would land on the Moon three years and four months later.
September and November 1966. The Programme Closes	43	Gemini 11 set an altitude record of 1,374 km that stood for decades. Gemini 12 closed the programme on 15 November 1966 with the first fully successful spacewalk, 2 hours 6 minutes, using handholds and rest periods designed after the failures of the previous three flights. Twenty months, ten crewed flights, every objective met. Nothing now stood between the agency and the Moon except a rocket, a lander, a suit, a map, and a fire.

The Smallest Spacecraft	44-45	One machine among eleven, treated as engineering and nothing else. The pressure suit was a one-man spacecraft: a wearable pressure vessel with a private atmosphere, private cooling, a private oxygen supply, and a helmet giving the wearer a narrow window onto the world and no peripheral vision at all. The suit had to be designed, built, tested, and requalified inside the same deadline as the rocket and the lander, and the suit was still being changed while the lander was being changed. The Apollo 1 investigation replaced the suit's outer layer with a non-flammable cloth.
The Suit's Schedule	46	A tipped-in card carrying only dates: when the suit was specified, when the suit was first delivered, when the suit was requalified after the fire, and when the lander the suit had to work inside was itself reconfigured. The two lists converge on one flight date and neither list slipped without moving the date.
PART IV. FIRE AND RECONNAISSANCE. 27 January 1967. Pad 34	47-48	A ground test, not a flight. Three crew: Grissom, White, Chaffee. A sealed cabin of pure oxygen at a pressure higher than sea level, running a rehearsal with the hatch closed. A spark in the wiring. An atmosphere that would not let anything burn slowly. A hatch that opened inward and would not open against cabin pressure. The programme lost three people in under a minute and had two years and eleven months of deadline left. The book does not moralise about this page. The book records what happened to the schedule next.

What Changed	49	The investigation produced changes, and the rebuild was completed inside the remaining deadline rather than by extending the deadline. The hatch was redesigned to open outward and blown in seconds. The cabin atmosphere at launch became 60% oxygen and 40% nitrogen instead of pure oxygen. Flammable materials were removed. Wiring and plumbing were rerouted and re-insulated. The suit's outer layer was replaced with a non-flammable cloth. This is the only section in the book listing what was changed instead of what was achieved.
The Address List	50-51	The Moon had to become a place with coordinates before anyone was sent to the Moon, and the mapping had to happen ahead of the crews rather than alongside the crews. Five Lunar Orbiters photographed 99% of the lunar surface between 1966 and 1967 at resolutions of 60 metres or better, ten times better than the best observation ever made from Earth, returning 1,654 photographs. Surveyor 1 soft-landed on 2 June 1966 and sent back 11,240 frames, the first showing the probe's own footpad resting on the soil. Across Ranger, Surveyor, and Lunar Orbiter, close to 100,000 photographs were taken to choose a landing site. The gatefold explains how a site was chosen: within a few degrees of the lunar equator, slope below a few degrees, reachable on a trajectory bringing the crew home even if the engine failed, lit by early-morning sun, and free of boulders.
The Question the Map Raises	52	One page, one question: Where do you land on a world with no addresses, when the address has to be chosen two years before anyone sees the world? The answer: you do not choose a place. You choose a set of constraints and let the constraints produce a place. Then you accept being four miles away from the place.

Rope Memory	53	The smallest machine in the book carried the largest risk. The guidance computer held 2,048 words of erasable core memory and 36,864 words of read-only memory woven by hand out of wire and magnetic cores, with 16-bit words. The computer was built at the MIT Instrumentation Laboratory, and the software team developed priority scheduling so the machine shed low-priority work under load and kept flying the vehicle. That single design decision is the reason the landing on p62 was not aborted. No claim is made anywhere in this book about what that work later did to consumer computing.
9 November 1967. The First One	54-55	The decision on pp 29 to 30 pays off. The first flight of the largest machine ever built carried all three live stages and a live spacecraft, on the first attempt: 363 feet tall, about 6.2 million pounds at liftoff, five first-stage engines producing 7.5 million pounds of thrust, one of the loudest sounds ever measured. The flight was nine and a half months after the fire. The third Saturn V ever launched carried people around the Moon. The sixth landed them.
11 October 1968. Twenty-Two Months Later	56	The rebuilt spacecraft flew with a crew for the first time: eleven days in Earth orbit, three men, every objective met. The hatch opened. The atmosphere held. Twenty months after the fire the machine worked, and the countdown read T minus 1y 02m 20d.

21 December 1968. Going Early	57	The boldest schedule change of the eleven years. With the deadline one year and ten days away, the landing itself six months and twenty-nine days away, and the lander still unproven with a crew aboard, the plan was changed so the next flight left Earth orbit entirely: three men to the Moon, ten orbits, and home. The change was decided in weeks and flown in months, and the change converted a spacecraft test into a rehearsal of everything except the landing. The Earthrise photographs from this flight are deliberately not in this book. The photographs belong to another student's topic. The flight is treated here only as a decision about sequence.
March 1969. The Lander Flies	58	The lander was flown in Earth orbit: undocked, separated, fired on the lander's own engine, rendezvoused, docked again, then jettisoned. Every function the lander would perform 240,000 miles away was performed once, with a crew, four months before the functions mattered.
May 1969. Fifty Thousand Feet	59	How close do you get before you dare to land? The dress rehearsal took the lander down to about 50,000 feet above the surface, photographed and radar-mapped the actual landing site, then flew back up, docked, and came home. Every step of the landing was rehearsed except the landing itself. Two months before the landing. Seven months before the deadline.
One Voice, One Clock	60	The organisation invented to run a machine at this speed. One controller per console. One shift per flight director. A rule: the flight director's decision was final for the duration of the shift. Only one person spoke to the crew at any time. Everything arrived at the room through one loop. Everything left through one mouth. The room had to be right the first time, because there was no second attempt inside the deadline.

The Board at Sixteen Days to Launch	61	The last page of Part IV and the densest page in the book: everything existing by 30 June 1969 in one board. Vehicle, crew, site, software, suit, tracking, recovery fleet, weather, and the eleven things that had to go right in the following fortnight.
PART V. TWELVE AND A HALF MINUTES. 20 July 1969. The Last Twelve and a Half Minutes	62	The final powered descent on one page: twelve and a half minutes, ending at 20:17:43 UTC. The automatic system was steering toward a boulder field beside a crater. The computer began raising overload alarms because a radar switch left in the wrong position was flooding the computer with interrupts, and the computer survived the alarms only because the computer had been designed to drop low-priority work and keep flying. The guidance controller in Houston, whose support engineer had seen the same alarm in a simulation, called go and the CAPCOM passed the call up. At about 300 feet the commander took manual control, flew at roughly 55 mph across the surface looking for clear ground, ignored a low-fuel warning, and touched down with about twenty seconds of descent propellant left, roughly four miles past the aim point.
20:17:43. Tranquility Base	63	Houston, Tranquility Base here. The Eagle has landed. The hatch opened about six and a half hours after touchdown. At 02:56 UTC on 21 July the first person stepped onto the surface. The walk lasted 2 hours 31 minutes, and 21.55 kilograms of rock and soil came home. An estimated 500 million people watched live. The surface stay was 21 hours 36 minutes.
The Third Man	64	Where is the third man? Alone in the orbiting spacecraft for thirty orbits, keeping station, photographing the surface, and passing behind the Moon out of all radio contact with anyone for nearly half of every revolution. The third man was the only one of the three who never set foot on the Moon, and for those intervals the third man was further from every other human being than anyone had ever been. This is the book's one page about a person rather than a machine.

21 July 1969, 17:54 UTC. Going Up	65	The ascent stage left the surface twenty-one and a half hours after the ascent stage arrived, on an engine that had to work once with nothing behind the engine. The engine had flown with a crew twice before, in Earth orbit and in lunar orbit, and this was the first time a way home depended on the engine. Rendezvous and docking followed within four hours. The lander was then jettisoned into lunar orbit. The commitment of May 1961 had two halves, and the second half, returning safely, is the half getting one page.
24 July 1969. Splashdown	66	Eight days, three hours, and eighteen minutes after liftoff, the Pacific. The countdown running head stops on this page at T minus 0y 05m 07d: the five months and seven days of the decade the landing never needed. The mission interval was 16 to 24 July 1969. The interval from the agency's first day to splashdown was ten years, nine months, and twenty-three days.
REPAIR / REPLACE / CARRY	67	The book's summary of how a compressed schedule handles failure, stated as three verbs and nothing else. REPAIR: fix in flight and keep going. The stuck thruster and the emergency abort of March 1966. The overload alarms waived by a go call in July 1969. REPLACE: rebuild the machine and the rules around the machine. The hatch, the cabin atmosphere, the wiring, and the suit after January 1967. CARRY: accept the fault and fly with the fault. Landing four miles long. Twenty seconds of propellant. A helmet giving ninety degrees of vision. A computer with 2,048 words of memory, rewritable. There is no fourth verb. There is no option to wait for more information.

After	68	The rate that never came back. Fiscal year 1966 was the peak at 4.41% of the federal budget and the share never returned. Today the share is a fraction of a percent. The last crewed Moon flight was December 1972. Two Saturn Vs were built and never flown. The eleven years were not a trend. The eleven years were a spike, produced by a fixed deadline, an unprecedented budget, a decision to run everything in parallel, and a tolerance for risk no peacetime organisation has accepted since. The book makes no argument about whether the spike was wise. The book argues only about what the spike took.	
The Acceleration Dial	69-70	The whole thesis in one object, with instructions and the eleven readings printed on p70. Turn the wheel to October 1958 and the four windows read: crew 0, time in space 0, distance from Earth 0, vehicle height nothing. Turn the wheel to July 1969 and the windows read: crew 3, time in space 8 days 3 hours, distance 384,400 km, vehicle 363 feet. The eleven intermediate positions are the eleven years, and every value on the dial is dated and sourced in section 6. The reader performs the acceleration with one hand.	